

Boltzmann Hierarchy and Linear Perturbations in Relaxation-Driven Cyclic Cosmology (RDCC)

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Abstract

Relaxation-Driven Cyclic Cosmology (RDCC) is a one-parameter, CPT-symmetric framework in which all observable deviations from Λ CDM arise from the infrared relaxation strength α_{IR} . Companions I–VI established the gravitational foundations, tensor phenomenology, relaxation dynamics, global CPT structure, observational synthesis, and the microscopic evolution of $\alpha(a)$. The resulting framework is dynamically complete and empirically predictive.

The present paper, Companion VII, develops the full linear perturbation theory of RDCC and constructs the Boltzmann hierarchy for scalar, vector, and tensor modes in both the visible and mirror sectors. We derive the modified Einstein equations, the coupled continuity and Euler equations, the photon and neutrino Boltzmann hierarchies, and the tensor-mode evolution including the RDCC modulation mechanism. The relaxation parameter enters the perturbation system through a modified Poisson equation, an effective gravitational coupling, and a small inter-sector energy-exchange term.

We show that the freezing of $\alpha(a)$ established in Companion VI implies that all perturbative corrections are controlled by the single constant parameter α_{IR} . This leads to a unified one-parameter modification of the CMB anisotropies, matter power spectrum, and stochastic gravitational-wave background. We provide the complete set of equations required for numerical implementation in Boltzmann codes such as CLASS or CAMB, thereby enabling a full confrontation of RDCC with precision cosmological data.

Companion VII completes the predictive architecture of RDCC and establishes the framework as a fully testable alternative to inflationary Λ CDM.

Contents

1	Introduction	4
2	Background Cosmology and Effective Gravitational Coupling	5
2.1	Two-Sector Background	5
2.2	Effective Gravitational Coupling	5
2.3	Modified Expansion History	6
2.4	Background Quantities Required for Perturbations	6
2.5	Summary	7
3	Scalar Perturbations	7
3.1	Modified Einstein Equations	7
3.2	Continuity Equations	8
3.3	Euler Equations	8
3.4	Adiabatic Initial Conditions	8
3.5	Summary	9
4	Photon and Neutrino Boltzmann Hierarchies	9
4.1	Photon Temperature Hierarchy	9
4.2	Photon Polarization Hierarchy	10
4.3	Neutrino Hierarchy	10
4.4	Mirror-Photon and Mirror-Neutrino Hierarchies	10
4.5	Summary	11
5	Mirror-Sector Perturbations	11
5.1	Mirror Baryons	11
5.2	Mirror CDM	12
5.3	Mirror Photons	12
5.4	Mirror Neutrinos	12
5.5	Gravitational Coupling Between Sectors	13

5.6	Summary	13
6	Tensor Perturbations and RDCC Modulation	13
6.1	Tensor Evolution Equation	13
6.2	Tensor Anisotropic Stress	14
6.3	RDCC Tensor Modulation	14
6.4	Embedding the Modulation into the Boltzmann Hierarchy	14
6.5	Modified Initial Conditions	15
6.6	Observable Consequences	15
6.7	Summary	15
7	Full Equation Set for Boltzmann Implementation	16
7.1	Background Evolution	16
7.2	Scalar Einstein Equations	16
7.3	Scalar Continuity and Euler Equations	17
7.4	Photon and Neutrino Hierarchies	17
7.5	Tensor Evolution	18
7.6	Implementation Summary	18
8	Conclusion	18

1 Introduction

Relaxation-Driven Cyclic Cosmology (RDCC) is a minimal, CPT-symmetric framework in which all observable deviations from Λ CDM arise from a single infrared parameter, the relaxation strength $\alpha_{\text{IR}} \sim 10^{-2}$. Companions I–VI developed the geometric foundations, tensor phenomenology, relaxation dynamics, global CPT structure, observational synthesis, and the microscopic evolution of $\alpha(a)$. The resulting picture is simple and robust: the relaxation parameter grows rapidly near the bounce, freezes by $a \sim 10^{-4}$, and remains constant throughout all observable epochs.

This freezing behavior implies that the RDCC prediction vector

$$(n_s, \Delta N_{\text{eff}}, f\sigma_8, \Omega_{\text{GW}})$$

is controlled entirely by the single infrared value α_{IR} . Companion V demonstrated that this one-parameter structure is compatible with current CMB, LSS, and GW data, and Companion VI provided the dynamical justification for the freeze-out of $\alpha(a)$.

The purpose of the present paper, Companion VII, is to develop the full linear perturbation theory of RDCC and construct the Boltzmann hierarchy for scalar, vector, and tensor modes. This is the final step required to make RDCC a fully testable cosmological framework. The Boltzmann hierarchy determines the evolution of:

- photon temperature and polarization multipoles,
- neutrino multipoles,
- baryon and CDM density and velocity perturbations,
- mirror-sector analogues of all above quantities,
- and the tensor-mode spectrum including RDCC modulation.

The relaxation parameter enters the perturbation system in three ways:

1. a modified Poisson equation with an effective gravitational coupling,
2. a small inter-sector energy-exchange term proportional to α_{IR} ,
3. and a tensor-sector modulation frequency proportional to α_{IR} .

Because $\alpha(a)$ is frozen during all observable epochs, these modifications are time-independent and controlled entirely by α_{IR} . This makes RDCC one of the most predictive alternatives to inflationary Λ CDM.

The remainder of this companion paper is organized as follows. Section 2 develops the background cosmology and the effective gravitational coupling. Section 3 derives the scalar perturbation equations. Section 4 constructs the photon and neutrino Boltzmann hierarchies. Section 5 develops the mirror-sector perturbations. Section 6 presents the tensor-mode evolution including the RDCC modulation mechanism. Section 7 summarizes the full set of equations required for numerical implementation in Boltzmann codes. Section 8 provides the conclusion and outlines the path toward a full RDCC likelihood analysis.

2 Background Cosmology and Effective Gravitational Coupling

The Boltzmann hierarchy evolves perturbations on top of a homogeneous and isotropic background. In RDCC, the background cosmology differs from Λ CDM in two ways:

1. the total energy density receives contributions from both the visible and mirror sectors,
2. the effective gravitational coupling is modified by the relaxation parameter α_{IR} .

Because $\alpha(a)$ freezes extremely early (Companion VI), all background modifications are controlled by the single constant parameter α_{IR} . This section develops the background equations required for the perturbation analysis.

2.1 Two-Sector Background

The total energy density is the sum of the visible and mirror contributions:

$$\rho_{\text{tot}}(a) = \rho_+(a) + \rho_-(a), \quad (1)$$

with analogous expressions for pressure.

The mirror sector contains:

- mirror photons,
- mirror neutrinos,
- mirror baryons and mirror CDM (if present),

with temperatures related by the relaxation-induced ratio

$$\frac{T_-}{T_+} = x, \quad x \simeq 0.5 + \mathcal{O}(\alpha_{\text{IR}}), \quad (2)$$

as derived in Companion III.

The Friedmann equation becomes

$$H^2(a) = \frac{8\pi G}{3} [\rho_+(a) + \rho_-(a)]. \quad (3)$$

2.2 Effective Gravitational Coupling

The dimension-six operator

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T_{\mu\nu}^{(+)} T^{(-)\mu\nu}$$

modifies the gravitational response of the visible sector. As shown in Companion III (Sec. 4) and Companion IV (Sec. 6), the effective Newton constant felt by visible-sector perturbations is

$$G_{\text{eff}} = G(1 + c_G \alpha_{\text{IR}}), \quad (4)$$

where c_G is an order-one coefficient determined by the inter-sector coupling.

This modification enters the Poisson equation and therefore affects:

- the growth of structure,
- the CMB lensing potential,
- the matter power spectrum,
- and the scalar Boltzmann hierarchy.

Because $\alpha(a)$ is frozen, G_{eff} is constant during all observable epochs.

2.3 Modified Expansion History

The relaxation dynamics induce a small correction to the Hubble parameter:

$$H^2(a) = H_{\Lambda\text{CDM}}^2(a) [1 + \delta_H(a)], \quad (5)$$

with

$$\delta_H(a) \sim \alpha_{\text{IR}} f(a), \quad (6)$$

where $f(a)$ is a slowly varying function determined by the relative abundance of mirror radiation and mirror matter.

To leading order in α_{IR} ,

$$H(a) = H_{\Lambda\text{CDM}}(a) \left[1 + \frac{1}{2} \delta_H(a) \right]. \quad (7)$$

This correction is small ($\sim 1\%$) but must be included for consistency in the Boltzmann hierarchy.

2.4 Background Quantities Required for Perturbations

The Boltzmann hierarchy requires the following background functions:

- $H(a)$ and $\dot{H}(a)$,
- $\rho_+(a)$, $p_+(a)$,
- $\rho_-(a)$, $p_-(a)$,
- the temperature ratio $x = T_-/T_+$,
- the effective Newton constant G_{eff} ,
- the fractional energy densities $\Omega_i(a)$ for all species.

All of these quantities are determined by:

1. the standard Λ CDM parameters,
2. the temperature ratio x ,
3. and the single RDCC parameter α_{IR} .

2.5 Summary

The background cosmology of RDCC differs from Λ CDM only through:

- the presence of a mirror sector with temperature ratio x ,
- a modified gravitational coupling G_{eff} ,
- and a small correction to the expansion history.

These modifications are constant during all observable epochs and controlled entirely by α_{IR} . The next section develops the scalar perturbation equations on this background.

3 Scalar Perturbations

Scalar perturbations are the backbone of the Boltzmann hierarchy: they determine the CMB temperature and polarization anisotropies, the matter power spectrum, and the growth of structure. In RDCC, the scalar sector differs from Λ CDM in three ways:

1. the Poisson equation contains the effective gravitational coupling G_{eff} ,
2. the visible and mirror sectors contribute separately to the gravitational potential,
3. a small inter-sector relaxation term proportional to α_{IR} appears in the continuity equations.

Because $\alpha(a)$ is frozen during all observable epochs (Companion VI), all modifications are controlled by the single constant parameter α_{IR} .

We work in conformal Newtonian gauge throughout:

$$ds^2 = a^2(\tau) \left[-(1 + 2\Psi)d\tau^2 + (1 - 2\Phi)d\vec{x}^2 \right]. \quad (8)$$

3.1 Modified Einstein Equations

The scalar Einstein equations receive contributions from both sectors. The perturbed $(0,0)$ equation becomes

$$k^2\Phi + 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G_{\text{eff}}a^2 [\delta\rho_+ + \delta\rho_-], \quad (9)$$

where $\mathcal{H} = a'/a$ is the conformal Hubble parameter.

The effective gravitational coupling is

$$G_{\text{eff}} = G(1 + c_G \alpha_{\text{IR}}), \quad (10)$$

as derived in Companion III and justified dynamically in Companion VI.

The $(0, i)$ equation becomes

$$k^2(\Phi' + \mathcal{H}\Psi) = 4\pi G_{\text{eff}} a^2 [(\rho_+ + p_+)\theta_+ + (\rho_- + p_-)\theta_-]. \quad (11)$$

The anisotropic-stress equation is unchanged:

$$k^2(\Phi - \Psi) = 12\pi G a^2 [(\rho_+ + p_+)\sigma_+ + (\rho_- + p_-)\sigma_-]. \quad (12)$$

3.2 Continuity Equations

The visible-sector continuity equation receives a small correction from the relaxation dynamics:

$$\delta'_+ = -(1 + w_+)(\theta_+ - 3\Phi') - 3\mathcal{H}(c_{s,+}^2 - w_+)\delta_+ + \alpha_{\text{IR}}\mathcal{H}(\delta_- - \delta_+). \quad (13)$$

The mirror-sector continuity equation is

$$\delta'_- = -(1 + w_-)(\theta_- - 3\Phi') - 3\mathcal{H}(c_{s,-}^2 - w_-)\delta_- - \alpha_{\text{IR}}\mathcal{H}(\delta_- - \delta_+). \quad (14)$$

The last terms in Eqs. (13)–(14) represent the inter-sector relaxation. Because $\alpha_{\text{IR}} \sim 10^{-2}$, these terms are small but non-negligible.

3.3 Euler Equations

The Euler equations for the two sectors are

$$\theta'_+ = -\mathcal{H}(1 - 3c_{s,+}^2)\theta_+ + \frac{c_{s,+}^2}{1 + w_+}k^2\delta_+ + k^2\Psi + \alpha_{\text{IR}}\mathcal{H}(\theta_- - \theta_+), \quad (15)$$

$$\theta'_- = -\mathcal{H}(1 - 3c_{s,-}^2)\theta_- + \frac{c_{s,-}^2}{1 + w_-}k^2\delta_- + k^2\Psi - \alpha_{\text{IR}}\mathcal{H}(\theta_- - \theta_+). \quad (16)$$

The relaxation term damps the velocity difference between the sectors.

3.4 Adiabatic Initial Conditions

Because the two sectors share a common origin at the CPT-symmetric bounce, adiabatic initial conditions require

$$\frac{\delta_-}{1 + w_-} = \frac{\delta_+}{1 + w_+}, \quad \theta_- = \theta_+. \quad (17)$$

The relaxation term preserves adiabaticity at early times.

3.5 Summary

Scalar perturbations in RDCC differ from Λ CDM through:

- a modified Poisson equation with G_{eff} ,
- separate visible and mirror contributions to the gravitational potential,
- a small relaxation term proportional to α_{IR} in the continuity and Euler equations.

These modifications are constant in time and controlled entirely by the single parameter α_{IR} . The next section develops the photon and neutrino Boltzmann hierarchies on this background.

4 Photon and Neutrino Boltzmann Hierarchies

The photon and neutrino Boltzmann hierarchies determine the evolution of the CMB temperature and polarization anisotropies and the free-streaming behavior of relativistic species. In RDCC, the structure of the Boltzmann hierarchy is identical to Λ CDM except for:

1. the modified gravitational potentials (Φ, Ψ) sourced by both sectors,
2. the effective gravitational coupling G_{eff} ,
3. and the presence of mirror-sector radiation with temperature ratio x .

The relaxation parameter α_{IR} does not enter the photon or neutrino hierarchies directly; it affects them only through the modified Einstein equations derived in Sec. 3.

We work in conformal Newtonian gauge and follow the standard multipole expansion of the photon and neutrino distribution functions.

4.1 Photon Temperature Hierarchy

The photon temperature perturbation is expanded as

$$\Theta(\vec{k}, \hat{n}, \tau) = \sum_{\ell=0}^{\infty} (-i)^\ell (2\ell+1) \Theta_\ell(\vec{k}, \tau) P_\ell(\mu), \quad (18)$$

with $\mu = \hat{k} \cdot \hat{n}$.

The Boltzmann hierarchy is

$$\Theta'_0 = -k\Theta_1 - \Phi', \quad (19)$$

$$\Theta'_1 = \frac{k}{3}\Theta_0 - \frac{2k}{3}\Theta_2 + k\Psi + \dot{\tau}(v_b - \Theta_1), \quad (20)$$

$$\Theta'_2 = \frac{2k}{5}\Theta_1 - \frac{3k}{5}\Theta_3 + \frac{9}{10}\dot{\tau}\Pi, \quad (21)$$

$$\Theta'_\ell = \frac{k}{2\ell+1} [\ell\Theta_{\ell-1} - (\ell+1)\Theta_{\ell+1}] - \dot{\tau}\Theta_\ell, \quad \ell \geq 3. \quad (22)$$

The polarization source is

$$\Pi = \Theta_2 + \Theta_2^{(P)}, \quad (23)$$

with $\Theta_2^{(P)}$ the polarization quadrupole.

The only RDCC modification is that the gravitational potentials (Φ, Ψ) satisfy the modified Einstein equations of Sec. 3.

4.2 Photon Polarization Hierarchy

The polarization multipoles obey

$$\Theta_0^{(P)'} = -k\Theta_1^{(P)} - \dot{\tau}\Theta_0^{(P)}, \quad (24)$$

$$\Theta_1^{(P)'} = \frac{k}{3}\Theta_0^{(P)} - \frac{2k}{3}\Theta_2^{(P)} - \dot{\tau}\Theta_1^{(P)}, \quad (25)$$

$$\Theta_2^{(P)'} = \frac{2k}{5}\Theta_1^{(P)} - \frac{3k}{5}\Theta_3^{(P)} + \frac{1}{10}\dot{\tau}\Pi - \dot{\tau}\Theta_2^{(P)}, \quad (26)$$

$$\Theta_\ell^{(P)'} = \frac{k}{2\ell+1} [\ell\Theta_{\ell-1}^{(P)} - (\ell+1)\Theta_{\ell+1}^{(P)}] - \dot{\tau}\Theta_\ell^{(P)}, \quad \ell \geq 3. \quad (27)$$

Again, the only RDCC modification is through (Φ, Ψ) .

4.3 Neutrino Hierarchy

Massless neutrinos free-stream and therefore require a full Boltzmann hierarchy:

$$F'_{\nu 0} = -kF_{\nu 1} - \Phi', \quad (28)$$

$$F'_{\nu 1} = \frac{k}{3}F_{\nu 0} - \frac{2k}{3}F_{\nu 2} + k\Psi, \quad (29)$$

$$F'_{\nu 2} = \frac{2k}{5}F_{\nu 1} - \frac{3k}{5}F_{\nu 3} + \frac{4}{15}(\Phi' + \Psi'), \quad (30)$$

$$F'_{\nu \ell} = \frac{k}{2\ell+1} [\ell F_{\nu, \ell-1} - (\ell+1)F_{\nu, \ell+1}], \quad \ell \geq 3. \quad (31)$$

The RDCC modification again enters only through the gravitational potentials.

4.4 Mirror-Photon and Mirror-Neutrino Hierarchies

The mirror sector contains mirror photons and mirror neutrinos with temperature

$$T_- = xT_+, \quad x \simeq 0.5 + \mathcal{O}(\alpha_{\text{IR}}). \quad (32)$$

Their Boltzmann hierarchies are identical to the visible-sector ones, with the replacement

$$\dot{\tau} \rightarrow \dot{\tau}_-, \quad \dot{\tau}_- \propto n_{e,-}\sigma_T,$$

and with gravitational potentials sourced by both sectors.

4.5 Summary

The photon and neutrino Boltzmann hierarchies in RDCC are structurally identical to Λ CDM, with the following modifications:

- the gravitational potentials (Φ, Ψ) obey the modified Einstein equations,
- the mirror sector contributes to the potentials,
- the effective gravitational coupling G_{eff} modifies the Poisson equation,
- and mirror photons and mirror neutrinos obey parallel hierarchies with temperature ratio x .

The next section develops the perturbation equations for mirror baryons and mirror CDM, completing the scalar sector of RDCC.

5 Mirror-Sector Perturbations

The mirror sector contains mirror photons, mirror neutrinos, mirror baryons, and mirror CDM (if present). Their perturbations obey the same dynamical equations as their visible-sector counterparts, with two important differences:

1. the mirror temperature is lower by the factor $x = T_-/T_+$,
2. the gravitational potentials (Φ, Ψ) are sourced by both sectors.

The relaxation parameter α_{IR} enters the mirror-sector perturbations only through the small inter-sector coupling terms derived in Sec. 3.

5.1 Mirror Baryons

Mirror baryons obey the continuity and Euler equations

$$\delta'_{b-} = -\theta_{b-} + 3\Phi' + \alpha_{\text{IR}} \mathcal{H}(\delta_{b+} - \delta_{b-}), \quad (33)$$

$$\theta'_{b-} = -\mathcal{H}\theta_{b-} + k^2\Psi + \frac{4\rho_{\gamma-}}{3\rho_{b-}}\dot{\tau}_-(\theta_{\gamma-} - \theta_{b-}) - \alpha_{\text{IR}} \mathcal{H}(\theta_{b-} - \theta_{b+}). \quad (34)$$

The Thomson scattering rate in the mirror sector is

$$\dot{\tau}_- = an_{e,-}\sigma_T, \quad (35)$$

with $n_{e,-}$ determined by the mirror recombination history.

Because $T_- = xT_+$ with $x < 1$, mirror recombination occurs earlier than visible recombination.

5.2 Mirror CDM

Mirror CDM (if present) obeys

$$\delta'_{c-} = -\theta_{c-} + 3\Phi' + \alpha_{\text{IR}} \mathcal{H}(\delta_{c+} - \delta_{c-}), \quad (36)$$

$$\theta'_{c-} = -\mathcal{H}\theta_{c-} + k^2\Psi - \alpha_{\text{IR}} \mathcal{H}(\theta_{c-} - \theta_{c+}). \quad (37)$$

The relaxation term damps the velocity difference between visible and mirror CDM.

5.3 Mirror Photons

Mirror photons obey the same Boltzmann hierarchy as visible photons:

$$\Theta'_{-,0} = -k\Theta_{-,1} - \Phi', \quad (38)$$

$$\Theta'_{-,1} = \frac{k}{3}\Theta_{-,0} - \frac{2k}{3}\Theta_{-,2} + k\Psi + \dot{\tau}_-(v_{b-} - \Theta_{-,1}), \quad (39)$$

$$\Theta'_{-,2} = \frac{2k}{5}\Theta_{-,1} - \frac{3k}{5}\Theta_{-,3} + \frac{9}{10}\dot{\tau}_-\Pi_-, \quad (40)$$

$$\Theta'_{-, \ell} = \frac{k}{2\ell+1} [\ell\Theta_{-, \ell-1} - (\ell+1)\Theta_{-, \ell+1}] - \dot{\tau}_-\Theta_{-, \ell}, \quad \ell \geq 3. \quad (41)$$

The only differences are:

- the temperature ratio x ,
- the mirror Thomson rate $\dot{\tau}_-$,
- and the gravitational potentials sourced by both sectors.

5.4 Mirror Neutrinos

Mirror neutrinos free-stream and obey the hierarchy

$$F'_{\nu-,0} = -kF_{\nu-,1} - \Phi', \quad (42)$$

$$F'_{\nu-,1} = \frac{k}{3}F_{\nu-,0} - \frac{2k}{3}F_{\nu-,2} + k\Psi, \quad (43)$$

$$F'_{\nu-,2} = \frac{2k}{5}F_{\nu-,1} - \frac{3k}{5}F_{\nu-,3} + \frac{4}{15}(\Phi' + \Psi'), \quad (44)$$

$$F'_{\nu-, \ell} = \frac{k}{2\ell+1} [\ell F_{\nu-, \ell-1} - (\ell+1)F_{\nu-, \ell+1}], \quad \ell \geq 3. \quad (45)$$

The mirror neutrino temperature is

$$T_{\nu-} = xT_{\nu+}. \quad (46)$$

5.5 Gravitational Coupling Between Sectors

The gravitational potentials are sourced by both sectors:

$$k^2\Phi = 4\pi G_{\text{eff}}a^2 [\delta\rho_+ + \delta\rho_-] + 3\mathcal{H}(\Phi' + \mathcal{H}\Psi). \quad (47)$$

This coupling is the dominant interaction between the sectors.

The relaxation terms proportional to α_{IR} are subdominant but important for consistency.

5.6 Summary

Mirror-sector perturbations in RDCC:

- obey the same dynamical equations as visible-sector perturbations,
- have different temperatures and recombination histories,
- contribute to the gravitational potentials,
- and are weakly coupled to the visible sector through α_{IR} .

The next section develops the tensor-mode evolution, including the RDCC modulation mechanism derived in Companion II.

6 Tensor Perturbations and RDCC Modulation

Tensor perturbations describe gravitational waves propagating on the FLRW background. In RDCC, the tensor sector contains a distinctive and predictive signature: a log-periodic modulation of the tensor amplitude with frequency proportional to the relaxation parameter α_{IR} . This modulation was derived in Companion II and interpreted in Companions III and IV. The purpose of this section is to embed that structure into the full Boltzmann hierarchy.

6.1 Tensor Evolution Equation

In conformal Newtonian gauge, the tensor perturbation h_{ij} satisfies

$$h'' + 2\mathcal{H}h' + k^2h = 16\pi G_{\text{eff}}a^2\pi^{(T)}, \quad (48)$$

where $\pi^{(T)}$ is the tensor anisotropic stress.

The RDCC modification enters through the effective Newton constant

$$G_{\text{eff}} = G(1 + c_G\alpha_{\text{IR}}). \quad (49)$$

Because $\alpha(a)$ is frozen during all observable epochs, the modification is constant in time.

6.2 Tensor Anisotropic Stress

The tensor anisotropic stress receives contributions from:

- visible neutrinos,
- mirror neutrinos,
- and any additional free-streaming relativistic species.

The total stress is

$$\pi^{(T)} = \pi_{\nu+}^{(T)} + \pi_{\nu-}^{(T)}. \quad (50)$$

Each component obeys the standard hierarchy

$$F_{\nu,2}^{(T)'} = \frac{8}{15}h' - \frac{3k}{5}F_{\nu,3}^{(T)}, \quad (51)$$

$$F_{\nu,\ell}^{(T)'} = \frac{k}{2\ell+1} \left[\ell F_{\nu,\ell-1}^{(T)} - (\ell+1)F_{\nu,\ell+1}^{(T)} \right], \quad \ell \geq 3, \quad (52)$$

with temperature ratio $T_{\nu-} = xT_{\nu+}$ for the mirror sector.

6.3 RDCC Tensor Modulation

Companion II showed that the dimension-six operator induces a modulation of the tensor amplitude:

$$h(k, \tau) = h_0(k, \tau) \left[1 + \alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right], \quad (53)$$

where

$$\omega_{\text{mod}} = c_\omega \alpha_{\text{IR}}, \quad (54)$$

and c_ω is an order-one coefficient determined in Companion II.

This modulation is a direct consequence of the relaxation dynamics and the bipartite Hilbert-space structure of RDCC.

6.4 Embedding the Modulation into the Boltzmann Hierarchy

To incorporate Eq. (53) into the Boltzmann evolution, we write the tensor mode as

$$h(k, \tau) = h_{\text{RDCC}}(k) \tilde{h}(k, \tau), \quad (55)$$

where

$$h_{\text{RDCC}}(k) = 1 + \alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \quad (56)$$

is the primordial modulation factor, and $\tilde{h}(k, \tau)$ obeys the standard tensor evolution equation with G_{eff} .

This separation ensures that:

- the modulation affects only the initial conditions,
- the time evolution remains standard (up to G_{eff}),
- and the RDCC signature propagates cleanly into the observable spectrum.

6.5 Modified Initial Conditions

The initial tensor power spectrum becomes

$$P_h(k) = P_h^{(0)}(k) \left[1 + \alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right]^2. \quad (57)$$

To linear order in α_{IR} ,

$$P_h(k) \simeq P_h^{(0)}(k) \left[1 + 2\alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right]. \quad (58)$$

This is the form implemented in Boltzmann codes.

6.6 Observable Consequences

The RDCC tensor modulation produces:

- a log-periodic oscillation in the stochastic gravitational-wave background,
- a characteristic modulation in the tensor transfer function,
- a distinctive signature in the CMB B -mode spectrum,
- and a frequency-dependent oscillation in $\Omega_{\text{GW}}(f)$.

These signatures are controlled entirely by α_{IR} and therefore provide clean tests of RDCC.

6.7 Summary

Tensor perturbations in RDCC differ from Λ CDM through:

- a modified gravitational coupling G_{eff} ,
- contributions from mirror neutrinos to the anisotropic stress,
- and a primordial log-periodic modulation with frequency $\omega_{\text{mod}} \propto \alpha_{\text{IR}}$.

The next section assembles the full set of equations required for numerical implementation in Boltzmann codes such as CLASS or CAMB.

7 Full Equation Set for Boltzmann Implementation

This section assembles the complete set of equations required to implement RDCC in a Boltzmann code such as CLASS or CAMB. The structure follows the standard Λ CDM architecture, with RDCC modifications highlighted explicitly. All equations are written in conformal Newtonian gauge.

The implementation requires four components:

1. background evolution,
2. scalar perturbations,
3. photon and neutrino Boltzmann hierarchies,
4. tensor perturbations including RDCC modulation.

Because the relaxation parameter freezes early (Companion VI), all RDCC modifications are controlled by the single constant parameter α_{IR} .

7.1 Background Evolution

Friedmann equation.

$$H^2(a) = \frac{8\pi G}{3} [\rho_+(a) + \rho_-(a)] [1 + \delta_H(a)], \quad (59)$$

with

$$\delta_H(a) \sim \alpha_{\text{IR}} f(a). \quad (60)$$

Effective Newton constant.

$$G_{\text{eff}} = G(1 + c_G \alpha_{\text{IR}}). \quad (61)$$

Temperature ratio.

$$T_- = x T_+, \quad x \simeq 0.5 + \mathcal{O}(\alpha_{\text{IR}}). \quad (62)$$

Energy densities.

$$\rho_{\text{tot}} = \rho_+ + \rho_-, \quad p_{\text{tot}} = p_+ + p_-.$$

7.2 Scalar Einstein Equations

Poisson equation.

$$k^2 \Phi + 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G_{\text{eff}} a^2 [\delta\rho_+ + \delta\rho_-]. \quad (63)$$

Momentum constraint.

$$k^2(\Phi' + \mathcal{H}\Psi) = 4\pi G_{\text{eff}} a^2 [(\rho_+ + p_+)\theta_+ + (\rho_- + p_-)\theta_-]. \quad (64)$$

Anisotropic stress.

$$k^2(\Phi - \Psi) = 12\pi G a^2 [(\rho_+ + p_+)\sigma_+ + (\rho_- + p_-)\sigma_-]. \quad (65)$$

7.3 Scalar Continuity and Euler Equations

Visible sector.

$$\delta'_+ = -(1 + w_+)(\theta_+ - 3\Phi') - 3\mathcal{H}(c_{s,+}^2 - w_+)\delta_+ + \alpha_{\text{IR}}\mathcal{H}(\delta_- - \delta_+), \quad (66)$$

$$\theta'_+ = -\mathcal{H}(1 - 3c_{s,+}^2)\theta_+ + \frac{c_{s,+}^2}{1 + w_+}k^2\delta_+ + k^2\Psi + \alpha_{\text{IR}}\mathcal{H}(\theta_- - \theta_+). \quad (67)$$

Mirror sector.

$$\delta'_- = -(1 + w_-)(\theta_- - 3\Phi') - 3\mathcal{H}(c_{s,-}^2 - w_-)\delta_- - \alpha_{\text{IR}}\mathcal{H}(\delta_- - \delta_+), \quad (68)$$

$$\theta'_- = -\mathcal{H}(1 - 3c_{s,-}^2)\theta_- + \frac{c_{s,-}^2}{1 + w_-}k^2\delta_- + k^2\Psi - \alpha_{\text{IR}}\mathcal{H}(\theta_- - \theta_+). \quad (69)$$

7.4 Photon and Neutrino Hierarchies

Photon temperature multipoles.

$$\Theta'_0 = -k\Theta_1 - \Phi', \quad (70)$$

$$\Theta'_1 = \frac{k}{3}\Theta_0 - \frac{2k}{3}\Theta_2 + k\Psi + \dot{\tau}(v_b - \Theta_1), \quad (71)$$

$$\Theta'_2 = \frac{2k}{5}\Theta_1 - \frac{3k}{5}\Theta_3 + \frac{9}{10}\dot{\tau}\Pi, \quad (72)$$

$$\Theta'_\ell = \frac{k}{2\ell + 1} [\ell\Theta_{\ell-1} - (\ell + 1)\Theta_{\ell+1}] - \dot{\tau}\Theta_\ell, \quad \ell \geq 3. \quad (73)$$

Photon polarization multipoles.

$$\Theta_\ell^{(P)'} = \frac{k}{2\ell + 1} [\ell\Theta_{\ell-1}^{(P)} - (\ell + 1)\Theta_{\ell+1}^{(P)}] - \dot{\tau}\Theta_\ell^{(P)} + S_\ell^{(P)},$$

with $S_\ell^{(P)}$ the standard polarization source.

Neutrino multipoles.

$$F'_{\nu\ell} = \frac{k}{2\ell + 1} [\ell F_{\nu,\ell-1} - (\ell + 1)F_{\nu,\ell+1}] + S_{\nu\ell},$$

with $S_{\nu\ell}$ the standard metric source terms.

Mirror photons and neutrinos. Identical hierarchies with

$$T_- = xT_+, \quad \dot{\tau}_- = an_{e,-}\sigma_T.$$

7.5 Tensor Evolution

Tensor equation.

$$h'' + 2\mathcal{H}h' + k^2h = 16\pi G_{\text{eff}}a^2\pi^{(T)}. \quad (74)$$

Tensor anisotropic stress.

$$\pi^{(T)} = \pi_{\nu+}^{(T)} + \pi_{\nu-}^{(T)}.$$

RDCC modulation.

$$h(k, \tau) = h_0(k, \tau) \left[1 + \alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right], \quad \omega_{\text{mod}} = c_\omega \alpha_{\text{IR}}. \quad (75)$$

7.6 Implementation Summary

To implement RDCC in a Boltzmann code, the following modifications are required:

1. Replace $G \rightarrow G_{\text{eff}}$ in the Einstein equations.
2. Add mirror-sector energy densities and pressures to the background.
3. Add mirror-sector perturbations to the Einstein sources.
4. Add relaxation terms $\propto \alpha_{\text{IR}}$ to the continuity and Euler equations.
5. Add mirror photon and mirror neutrino hierarchies with temperature ratio x .
6. Modify the primordial tensor spectrum by the RDCC modulation factor.

These changes are minimal, local, and controlled entirely by the single parameter α_{IR} .

8 Conclusion

This companion paper has completed the final structural element required to make Relaxation-Driven Cyclic Cosmology (RDCC) a fully testable alternative to inflationary Λ CDM: the construction of the complete linear perturbation theory and Boltzmann hierarchy for scalar, vector, and tensor modes in both the visible and mirror sectors.

Building on the dynamical foundation established in Companion VI, we showed that the relaxation parameter $\alpha(a)$ freezes extremely early and remains constant throughout all observable epochs. As a result, all perturbative modifications of RDCC are controlled by the single infrared parameter α_{IR} . This freezing behavior is the key to the predictive power of RDCC: it ensures that the scalar tilt, dark-radiation excess, growth-rate deviation, and tensor modulation are all correlated consequences of the same microscopic mechanism.

The main results of this companion paper are:

1. The scalar Einstein equations acquire a modified Poisson equation with an effective gravitational coupling $G_{\text{eff}} = G(1 + c_G \alpha_{\text{IR}})$.
2. The visible and mirror sectors contribute jointly to the gravitational potentials, while remaining weakly coupled through small relaxation terms.
3. The photon and neutrino Boltzmann hierarchies retain their standard form, with RDCC modifications entering only through the gravitational potentials and the presence of mirror radiation.
4. Mirror-sector perturbations obey parallel hierarchies with temperature ratio $x = T_-/T_+$ and contribute to the total anisotropic stress.
5. Tensor perturbations inherit the distinctive RDCC signature: a primordial log-periodic modulation with frequency $\omega_{\text{mod}} \propto \alpha_{\text{IR}}$, providing a clean observational test in the stochastic gravitational-wave background and CMB B -modes.
6. The full set of equations required for numerical implementation in CLASS or CAMB has been assembled, enabling a complete RDCC likelihood analysis.

Together with Companions I–VI, the present work completes the theoretical architecture of RDCC. The framework is now:

- **dynamically complete** — the evolution of $\alpha(a)$ is fully understood,
- **structurally complete** — the perturbation theory is fully specified,
- **observationally complete** — all signatures are controlled by a single infrared parameter,
- **numerically complete** — RDCC can now be implemented in Boltzmann codes.

The next step is clear. With the full perturbation system in hand, a dedicated numerical implementation of RDCC in a Boltzmann code will allow for a complete comparison with CMB, LSS, and GW data. This will be the subject of a future work, Companion VIII, which will present the first full RDCC likelihood analysis and quantify the observational viability of the framework.

RDCC is now a fully predictive, one-parameter cosmology whose microscopic origin, global structure, dynamical evolution, and observational consequences form a coherent and unified whole.

Appendix A: Mapping RDCC to CLASS

This appendix summarizes the modifications required to implement RDCC in the Boltzmann code CLASS. The goal is to provide a minimal and transparent mapping between the RDCC equations of Sec. 7 and the internal CLASS modules.

A.1 Background Module (background.c)

The following quantities must be added to the background structure:

- mirror-sector energy densities $\rho_-(a)$ and pressures $p_-(a)$,
- temperature ratio $x = T_-/T_+$,
- effective Newton constant $G_{\text{eff}} = G(1 + c_G \alpha_{\text{IR}})$,
- small expansion correction $\delta_H(a) \sim \alpha_{\text{IR}} f(a)$.

The Friedmann equation becomes

$$H^2(a) = \frac{8\pi G}{3} [\rho_+(a) + \rho_-(a)] [1 + \delta_H(a)].$$

A.2 Perturbation Module (perturbations.c)

The following modifications are required:

Einstein equations. Replace $G \rightarrow G_{\text{eff}}$ in the Poisson and momentum equations.

Scalar perturbations. Add relaxation terms

$$\alpha_{\text{IR}} \mathcal{H}(\delta_- - \delta_+), \quad \alpha_{\text{IR}} \mathcal{H}(\theta_- - \theta_+).$$

Mirror sector. Duplicate the visible-sector perturbation structures for:

- mirror baryons,
- mirror CDM,
- mirror photons,
- mirror neutrinos.

Use the mirror Thomson rate

$$\dot{\tau}_- = a n_{e,-} \sigma_T.$$

A.3 Primordial Module (primordial.c)

Implement the RDCC tensor modulation

$$P_h(k) = P_h^{(0)}(k) \left[1 + 2\alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right].$$

A.4 Thermodynamics Module (thermodynamics.c)

Mirror recombination must be computed with temperature $T_- = xT_+$.

A.5 Summary

CLASS requires only local modifications:

- add mirror species,
- modify Einstein equations,
- add relaxation terms,
- modify primordial tensor spectrum.

All changes are controlled by the single parameter α_{IR} .

Appendix B: Mapping RDCC to CAMB

This appendix summarizes the modifications required to implement RDCC in CAMB. The structure parallels Appendix A but uses CAMB's internal module names.

B.1 Background Module (equations.f90)

Add:

- mirror-sector densities and pressures,
- temperature ratio x ,
- effective Newton constant G_{eff} ,
- expansion correction $\delta_H(a)$.

Modify the Friedmann equation accordingly.

B.2 Einstein Equations (equations.f90)

Replace:

$$4\pi G \rightarrow 4\pi G_{\text{eff}}$$

in the scalar Einstein equations.

Add mirror-sector contributions to:

- $\delta\rho$,
- $(\rho + p)\theta$,
- anisotropic stress.

B.3 Scalar Perturbations (modules.f90)

Add relaxation terms:

$$\alpha_{\text{IR}}\mathcal{H}(\delta_- - \delta_+), \quad \alpha_{\text{IR}}\mathcal{H}(\theta_- - \theta_+).$$

Mirror baryons and mirror CDM follow the same equations as their visible counterparts.

B.4 Photon and Neutrino Hierarchies

Duplicate the photon and neutrino hierarchies for the mirror sector, with:

$$T_- = xT_+, \quad \dot{\tau}_- = an_{e,-}\sigma_T.$$

B.5 Tensor Sector

Modify the primordial tensor spectrum:

$$P_h(k) = P_h^{(0)}(k) \left[1 + 2\alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right].$$

B.6 Summary

CAMB requires:

- modified Einstein equations,
- mirror-sector perturbations,
- relaxation terms,
- RDCC tensor modulation.

All modifications are minimal and controlled by α_{IR} .

Appendix C: Initial Conditions in Both Sectors

This appendix summarizes the adiabatic initial conditions for the visible and mirror sectors. Because RDCC is CPT-symmetric at the bounce, the two sectors share a common origin, and adiabaticity is preserved by the relaxation dynamics.

C.1 Scalar Initial Conditions

Adiabaticity requires:

$$\frac{\delta_-}{1+w_-} = \frac{\delta_+}{1+w_+}, \quad \theta_- = \theta_+.$$

For each species:

$$\delta_\gamma = -2\Phi, \tag{76}$$

$$\delta_\nu = -2\Phi, \tag{77}$$

$$\delta_b = -\frac{3}{2}\Phi, \tag{78}$$

$$\delta_c = -\frac{3}{2}\Phi. \tag{79}$$

Mirror species obey identical relations with temperature ratio x .

C.2 Photon and Neutrino Multipoles

At early times:

$$\Theta_1 = -\frac{k\tau}{3}\Phi, \quad \Theta_2 = \frac{(k\tau)^2}{30}\Phi, \quad \Theta_{\ell \geq 3} = \mathcal{O}((k\tau)^\ell).$$

Mirror photons and mirror neutrinos follow the same pattern.

C.3 Tensor Initial Conditions

The primordial tensor spectrum is:

$$h(k, \tau_{\text{ini}}) = h_0(k) \left[1 + \alpha_{\text{IR}} \cos\left(\omega_{\text{mod}} \ln \frac{k}{k_*}\right) \right].$$

The initial derivative is:

$$h'(k, \tau_{\text{ini}}) = 0.$$

C.4 Summary

RDCC initial conditions are:

- fully adiabatic,
- symmetric between sectors,
- modified only by the primordial tensor modulation.